Omar Abo-Elmaaty

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About Me

Full-Stack Engineer building production AI-powered applications across frontend interfaces and backend services. Currently at GRINDA AI, developing end-to-end features with React, Next.js, and TypeScript, spanning API integrations, authentication, and AI agent workflows. Combines strong UI/UX skills with backend engineering to deliver reliable, maintainable applications and responsive user experiences.

Education

Alexandria University Faculty Of Science — Alexandria, Egypt

Bachelor of Science in Software Industry and Multimedia // GPA: 3 (Very Good) — *Aug. 2020 – July 2024*

Experience

Full-Stack Engineer — GRINDA AI — Seoul, South Korea — Remote — Aug 2024 – Present

- Develop and maintain full-stack features for AI-powered applications, connecting React/Next.js interfaces with backend services and third-party APIs.
- Build AI agent integrations and application workflows, handling data exchange, execution states, and errors across the stack.
- Implement secure authentication flows and protected routes across application interfaces and backend endpoints.
- Strengthen AI agent execution security through sandbox network restrictions and server-side credential handling.
- Create reusable TypeScript components and responsive interfaces, translating product requirements into maintainable application features.
- Optimize frontend performance through lazy loading and code splitting, and investigate production issues across frontend and backend workflows.

Front-End Development Intern — GRINDA AI — Seoul, South Korea — Remote — Jul 2024 – Aug 2024

- Developed responsive interfaces and interactive components for two AI-powered applications using React and Next.js.
- Worked with Supabase authentication and Next.js server-side rendering, collaborating with senior developers to improve usability and resolve interface issues.

Projects

Wimmo — Focus App (Live on Google Play) [Preview] — React Native, Expo, TypeScript, AWS, Gemini, Reven

- Shipped an offline-first React Native app to Google Play with on-device SQLite and English/Arabic RTL.
- Built a Gemini (Vertex AI) planner behind an AWS Lambda proxy that turns a vague task into small first steps, with a validator that rejects malformed plans.
- Integrated Google Play Billing and RevenueCat, with access decided on the server from verified store webhooks.
- Built the AWS backend (Lambda, DynamoDB, Cognito, SES) with 182 server tests and a daily ops email of installs, crashes and AI cost.

Arator — AI Employee Platform [Preview] — Next.js, TypeScript, Tailwind CSS, TanStack Query, PostgreSQL

- Built live run observability: real-time call counts, last-event age, and a stall detector that tells a slow run from a hung one.
- Redesigned the run page into a two-column layout and folded the agent's sidebar block into the team roster with live presence.
- Tripled above-the-fold density on the daily brief by rebuilding its type scale and spacing, keeping EN/KO copy intact.
- Mapped raw harness errors to human, actionable failure messages, shipped through mockup-first approvals and a 760-test CI suite.

Opener — Autonomous Reddit Marketing Agent [Preview] — Next.js, TypeScript, Tailwind CSS, Prisma, Postg

- Built a Reddit marketing agent that finds buyers, scores them with a two-tier Claude pipeline for buying intent and product fit, and drafts replies that wait in an approval queue so nothing posts without a human.
- Made Reddit reads survive network blocking with a fallback ladder — OAuth, then public JSON, then RSS, then the browser — so collection keeps working when any one host cuts us off.
- Added an account-health dashboard that spots shadowbans from RSS visibility, tells them apart from real bans, and counts removals over a true rolling 7 days.
- Turned each subreddit's rules into a machine-checked ban list and a disclosure gate, so a draft that breaks a rule is stopped before it ever posts.

HIRAKU Agent [Preview] — Next.js, TypeScript, Tailwind CSS, Prisma, PostgreSQL (Supabase), Playwright,

- Built an AI agent that fills and submits Japanese contact forms end to end, using Playwright with an LLM mapper and deterministic fallbacks so it works even without an AI key.
- Added a learning layer on Prisma/Postgres so the agent remembers answers to fields it couldn't fill — and stops asking after the first time.
- Developed the Next.js dashboard and marketing site in TypeScript + Tailwind, localized in ja / en / ko, with a WebGL shader hero and an animated campaign wizard.
- Hardened the send path with SSRF protection, compliance checks (no-solicitation, CAPTCHA → human), and 340 unit tests.

Excel-Smart [Preview] — Next.js, FastAPI, TypeScript, Anthropic Claude, Tailwind CSS, Better AuthJun 2026

- Built a bilingual (Arabic/English) SaaS that turns messy Excel workbooks into clean, verified accounting systems.
- Engineered an AI pipeline on the Claude API that structures uploaded spreadsheets under a human-in-the-loop approval flow.
- Implemented multi-tenant auth with Better Auth and signed JWT assertions, enforcing strict per-tenant data isolation.
- Hardened .xlsx ingestion with server-side preflight validation, plus Excel export and PDF statement generation.
- Crafted an accessible UI (full RTL, dark mode, keyboard command palette) and shipped with Redis-backed rate limiting and Sentry monitoring.

Trade Show Agent [Preview] — React.js, Tailwind CSS, Supabase, Context API Jan 2025

- Architected a user-friendly and highly responsive web interface optimized for efficient trade show preparation and sales strategy creation.
- Implemented secure user authentication and management features utilizing Supabase integration.
- Enhanced user experience and perceived performance by implementing seamless dynamic loading states and smooth UI transitions during asynchronous data processing.

Reconstruction-Copilot — Next.js, Tailwind CSS, CopilotKit, TypeScript Nov 2024

- Recreated a high-fidelity front-end UI by precisely replicating complex Figma designs (Wooriga OMS) ensuring strict visual consistency and responsive layouts.
- Integrated AI capabilities into a custom Chat Bubble UI using CopilotKit to enable real-time, interactive, and context-aware assistance.
- Enhanced platform functionality by seamlessly embedding Copilot features across multiple modules for a unified user experience.

BNK-BAIA Agent [Preview] — React.js, Redux Toolkit, Tailwind CSS Nov 2024

- Developed a responsive AI assistant interface for financial document analysis, report generation, and code suggestions, with streaming responses and caching.
- Built conversation history and knowledge base interfaces for organizing chats, uploading financial documents, and managing regulatory updates.
- Created an admin dashboard with secure authentication, live user activity monitoring, and system performance analytics.

Compliance-AI [Preview] — Next.js, Redux Toolkit, i18next, Axios Oct 2024

- Built AI-powered document analysis interfaces integrated with backend APIs to automate compliance checks.
- Developed compliance gap analysis and document lookup features to identify missing ISO requirements and reference relevant standards.
- Implemented multilingual interfaces with i18next, managed application state with Redux Toolkit, and added authentication flows with protected routes.

Skills

Core Languages: JavaScript (ES6+), TypeScript, HTML5, CSS3, Python, SQL (PostgreSQL)
Frontend Frameworks: React.js, Next.js (SSR/SSG), Redux Toolkit, Context API, Tailwind CSS, jQuery
Backend / APIs: Node.js, Express.js, Flask, FastAPI, RESTful APIs, GraphQL, tRPC
Auth & Database: Supabase, Firebase, JWT, Protected Routes, PostgreSQL
Developer Tools: Git, Docker, Linux, VS Code, i18next, CopilotKit
Other Languages: Java, C++
Soft Skills: Problem-Solving, Cross-functional Collaboration, Adaptability, Time Management
Languages: Arabic (Native), English (Proficient)